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[5152]-108

S.E. (Civil Engineering) (II Sem.) EXAMINATION, 2017

CONCRETE TECHNOLOGY

(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Answer Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4, Q. No. 5 or Q. No. 6, Q. No. 7 or Q. No. 8.

(ii) Neat diagrams must be drawn wherever necessary.

(iii) Figures to the right indicate full marks.

(iv) Your answers will be valued as a whole.

(v) Use of electronic pocket calculator is allowed.

(vi) Assume suitable data, if necessary.

(vii) Use of IS code 10262, 456 is not allowed.

1. (a) Explain wet process of manufacturing of Portland cement. [6]
(b) Define workability of concrete. Explain the factors affecting workability. [6]

Or

2. (a) Write a short note on classification of aggregates. [6]
(b) Write short notes on : [6]
(i) Shrinkage
(ii) Swelling.

P.T.O.

3. (a) What are the test cores ? What are the advantages and disadvantages of test cores ? [6]
- (b) Describe the different types of vibrators used for compaction of concrete. [6]

Or

4. (a) Write short notes on : [6]
- (i) Light weight concrete
- (ii) Under water concreting
- (b) Write the comparison between Ferro cement and RCC construction work. [6]
5. (a) What do you mean by concrete mix design ? What are the objectives in mix design ? [6]
- (b) Explain mix design by IS recommended guidelines in detail. [7]

Or

6. (a) Explain the factors affecting the mix design. [4]
- (b) Write a short note on statistical quality control of concrete. [4]
- (c) Briefly outline DOE method of mix design. [5]
7. (a) Explain in detail permeability and factors affecting permeability of the concrete. [5]

(b) Write short notes on : [8]

(i) Corrosion monitoring techniques and its preventive measures

(ii) Shotcrete.

Or

8. (a) What is evaluation of cracks ? Why is it necessary ? [5]

(b) Write short notes on : [8]

(i) Attack by sea water

(ii) Carbonation of concrete and its determination.